

Factoring trinomials



1

a No

b Yes

c Yes

d No

2

a $(5 + v)(a + b)$

b $(y - z)(x - w)$

c $(2y + 3w)(8x - z)$

d $(5x^2 + 4y)(8z - 1)$

e $2(y - 4)(4y - 5)$

f $(4c - d)(c + 9d + 5)$

3

a $(x + 5)(x + 8)$

b $(x - 5)(x + 10)$

4

a $(x + 2)(y - 9)$

b $(8 - x)(7 - y)$

c $(x + z)(12y + w)$

d $(x - 20y)(2 + z)$

e $(5 + x)(10 + y)$

f $(12y + w)(x + z)$

5 $(x + 4)(4x + 1)$

6 To factor the quadratic $x^2 + 11x + 24$, we need to find two numbers whose product is 24 and whose sum is 11.

7

a $a = 7, b = 8$

b $a = -10, b = -5$

8

a $a = -2, b = -9$

b $a = 8, b = -14$

9

a $m + 10$

b $y + 5$

10 $(m + 5)(m + 9)$

11

a $(x + 5)(x + 4)$

b $(3x + 1)(x + 6)$

12

a $(x + 1)(x + 5)$

b $(x + 4)(x + 10)$



c $(x - 1)(x - 4)$

d $(x - 3)(x - 10)$

e $(x - 8)(x - 7)$

f $(x + 12)(x - 11)$

g $(x + 15)(x - 6)$

h $(x + 2)(x - 4)$

i $(x + 7)(x - 10)$

j $(x + 2)(x + 3)$

k $(x - 11)(x - 4)$

l $-(x + 1)(x + 12)$

m $-(x - 9)(x - 1)$

13

a $4(x + 6)(x + 8)$

b $-5(x - 4)(x + 2)$

c $3(x - 9)(x + 2)$

d $2x(x + 5)(x + 3)$

14 $-10(t + 2)(t - 7)$

15

a $-16(t + 1)(t - 2)$

b 0

c 0

d $-16t^2 + 16t + 32$ and $-16(t + 1)(t - 2)$ evaluated at $t = 2$ are both equal to 0.
This means that 2 seconds after jumping off the board, Tara has hit the water.

16 $(4 + x)(3 - x)$ square units

17

a $3x^2 + 10x - 8 = (3x - 2)(x + 4)$

b $12x^2 + 25x + 7 = (4x + 7)(3x + 1)$

c $15x^2 - 13x - 6 = (5x - 6)(3x + 1)$

18

a $(7x - 5)(x - 10)$

b $(6x + 7)(x - 1)$

c $(9x - 10)(x - 1)$

d $(7x + 8)(x - 6)$

e $(3x + 2)(2x + 3)$

f $-(4x - 3)(3x + 4)$

g $(9 - 8x)(10 - x)$ or $(8x - 9)(x - 10)$

h $-(7p - 2)(7p + 4)$

19 $2x + 5, 3x + 4$

20

a $10y^2 - 112y + 22$

b $2(5y - 1)(y - 11)$

21 9, -9, 15, -15



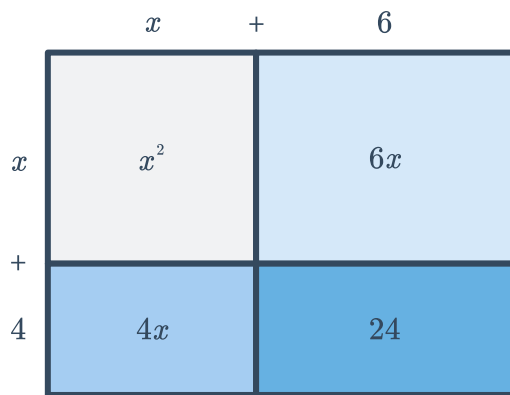
Additional questions

23

- a $(x - 6)(x + 7)$
 b $x^2 + x - 42$
 c The sum of 7 and -6 is 1. The product of 7 and -6 is -42 .

24

a

b $(x + 6)(x + 4)$

25

- a Exactly one of p or q must be negative.
 b Both p and q must be positive.
 c Both p and q must be negative.

26

- a $(x + 9)(x + 8) = x^2 + 17x + 72$
 b $(x + 2)(x + 1) = x^2 + 3x + 2$
 c $(x + 1)(x - 9) = x^2 - 8x - 9$

27

- a To maximize the length of the entire field you would increase the length by 36 and the width by 1
 b If both portions are extended by 6 then the areas in the bottom left and top right will be the same, which would minimize the difference in area.

28

- a 15 b The same c -16 d Opposites

29

a -45

b Opposites

c -4

d Opposites

30 Although this expression appears to have a leading coefficient of 3, the 3 is a common factor that can be factored out of the expression which leaves a trinomial where $a = 1$. The fully factored form is $3(x^2 + 5x - 21) = 3(x + 7)(x - 5)$.

31

a Answers may vary.

 $4x^2 + 12x + 5 = (2x + 1)(2x + 5)$ is one possible answer.

32

a Each polynomial has a leading coefficient $a \neq 1$ and none of the polynomials can be factored.

b Find the product $a \times c$, then look for factors of ac that can add or subtract to be equal to the value of b . If no such factors exist, the polynomial is not factorable.

33

a She would use x to represent the width of her border if she did not know yet what size border she wanted.

b The expression for area factors as $(2x + 5)(2x + 7)$ so the dimensions of the photograph are $2x + 5$ and $2x + 7$

c If the border is a width of x you would subtract $2x$ from each dimension since there is a border on both sides. That leaves the dimensions as 3 and $2x - 1$ so the area of the photograph is $6x - 3$ square units.

34 Answers may vary.

In both cases we are looking for a pair of binomials where the leading coefficients of the factors will multiply to be a and the constant terms will multiply to be c . The difference is that in a polynomial where $a \neq 1$ the factors of the constant c won't add up to the coefficient of the linear term, b .

35

a The largest possible value for b is 71 and the factors are $(12x - 1)(x + 6)$

b The smallest possible value for b is 1 and the factors are $(4x + 3)(3x - 2)$.